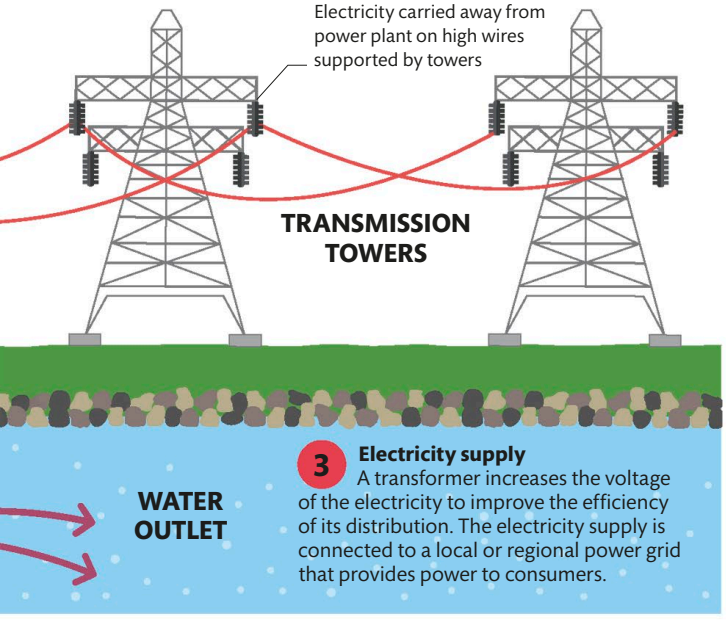
THE DANGERS OF DRILLING

Enhanced geothermal systems (EGSs) inject fluid under high pressure to create cracks and fractures in rock so that the fluid can travel through a larger area and obtain more heat. There is some evidence that such fracturing could create uncontrollable seismic activity. In 2006, a geothermal plant in Basel, Switzerland, was held responsible for inducing a magnitude 3.4 earthquake. Eleven years later, a magnitude 5.4 earthquake in Pohang, South Korea, injured 82 people. Initial studies suggest a local geothermal plant may have been the cause.

THE **ITAIPU DAM** ON THE PARAGUAY–BRAZIL BORDER PROVIDES **76 PERCENT** OF **PARAGUAY'S ENERGY**



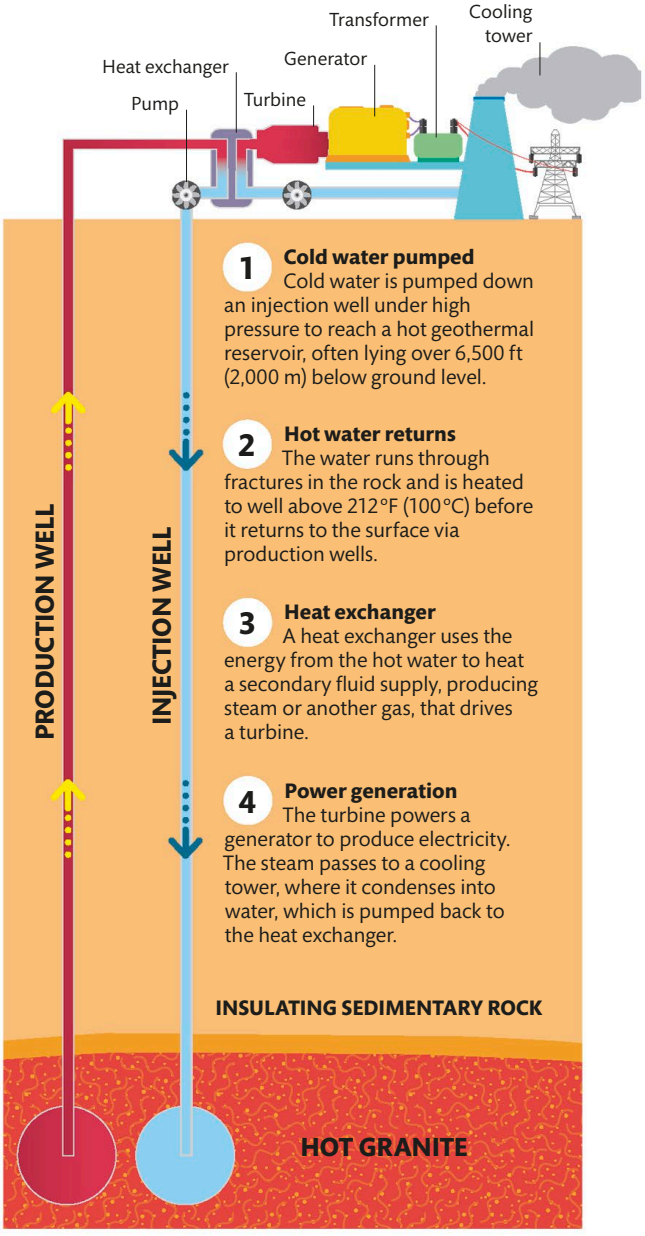
Channeling water
A continuous strong flow of water is needed for an HEP scheme to generate electricity constantly. Some schemes, known as pumped storage HEP systems, pump the outlet water back up to the reservoir at times of lower electricity demand, using surplus electricity.



3 Electricity supply
A transformer increases the voltage of the electricity to improve the efficiency of its distribution. The electricity supply is connected to a local or regional power grid that provides power to consumers.

Geothermal power

Heat from hot underground rocks can be harnessed in different ways. Underground water can be tapped directly, or water can be pumped through a geothermal region to gain heat used to generate electricity. Geothermal power produces a small fraction of the harmful emissions of a coal-fired power station.



1 Cold water pumped
Cold water is pumped down an injection well under high pressure to reach a hot geothermal reservoir, often lying over 6,500 ft (2,000 m) below ground level.

2 Hot water returns
The water runs through fractures in the rock and is heated to well above 212°F (100°C) before it returns to the surface via production wells.

3 Heat exchanger
A heat exchanger uses the energy from the hot water to heat a secondary fluid supply, producing steam or another gas, that drives a turbine.

4 Power generation
The turbine powers a generator to produce electricity. The steam passes to a cooling tower, where it condenses into water, which is pumped back to the heat exchanger.